

Chapter 6: Chemical Formulae and Equations

Chemical Formulae of Common Acids

Hydrochloric Acid $\rightarrow$ Hydrogen Chloride $\rightarrow$ HCl

Sulfuric Acid $\rightarrow$ Hydrogen Sulfate $\rightarrow$ H_2SO_4

Nitric Acid $\rightarrow$ Hydrogen Nitrate $\rightarrow$ HNO_3

Carbonic Acid $\rightarrow$ Hydrogen Carbonate $\rightarrow$ H_2CO_3

Phosphoric Acid $\rightarrow$ Hydrogen Phosphate $\rightarrow$ H_3PO_4

Covalent Substances - How to deduce formula

Prefix Examples:

prefix	number of atoms	Examples:
mono-	1	Dinitrogen tetroxide $\rightarrow \text{N}_2\text{O}_4$
di-	2	Carbon monoxide $\rightarrow \text{CO}$
tri-	3	(note: if the first element is single, mono- is omitted.)
tetra-	4	

Covalent compounds to memorise (cannot deduce from words)

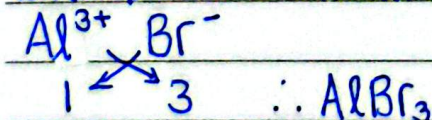
1. Water - H_2O
2. Methane - CH_4
3. ammonia - NH_3
4. hydrogen peroxide - H_2O_2

Ionic Compounds - How to deduce formula

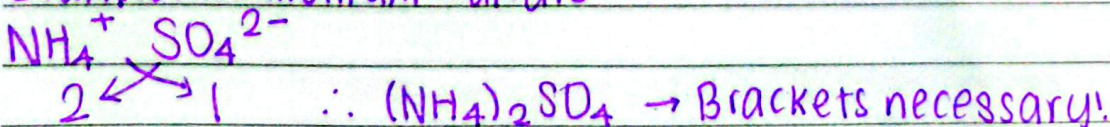
Step 1: Identify metal and non-metal and their charge.

Step 2: Cross multiply to deduce formula. (Simplify when possible)

Example: Aluminium Bromide



Example: Ammonium Sulfate



Valency

The valency of an element can be referred to as the number of electrons that must be lost, gained or shared in order for that element to obtain a stable, noble gas electronic configuration.

The valency of an ion is its charge without the sign.

Valency From Group Numbers

Group Number	Loss/Gain/Share	Valency	Formula of Ion
1	Lose 1 electron	1	+
2	Lose 2 electrons	2	2+
13	Lose 3 electrons	3	3+
14	Share 4 electrons	4	
15	Gain 3 electrons	3	3-
16	Gain 2 electrons	2	2-
17	Gain 1 electron	1	-
18	—	Zero valency	

Valencies from Roman Numerals (Transition Metals)

I $\rightarrow$ 1 Examples: copper(I) $\rightarrow$ Cu^+

II $\rightarrow$ 2 copper(II) $\rightarrow$ Cu^{2+}

III $\rightarrow$ 3

IV $\rightarrow$ 4

Special cases: Silver and zinc

Silver and zinc have fixed valencies: Ag^+ and Zn^{2+}

Polyatomic Ions

- Polyatomic ions are made of more than one atom covalently bonded together.

Examples of Polyatomic Ions

1. hydroxide $\rightarrow \text{OH}^-$
2. nitrate $\rightarrow \text{NO}_3^-$
3. sulfate $\rightarrow \text{SO}_4^{2-}$
4. carbonate $\rightarrow \text{CO}_3^{2-}$
5. ammonium $\rightarrow \text{NH}_4^+$
6. phosphate $\rightarrow \text{PO}_4^{3-}$

Balancing Chemical Equations

Step 1: Identify reactants (LHS) and products (RHS)

Step 2: Identify the chemical formula of reactants and products

Step 3: State symbols (if necessary)

Step 4: Balance the chemical equation (below)

Step 4: How to balance the chemical equation

1. Place a letter (a, b, c, d...) in front of every reactant/product
2. For each element, write an equation in terms of a, b, c, d...
based on how many ~~elements~~ atoms there are on each side
3. Let $a = 1$ then find the rest of the variables.

Example: Copper + Nitric Acid $\rightarrow$ Copper(II) Nitrate + Nitrogen Dioxide + water

Step 2: $\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2 + \text{H}_2\text{O}$

Step 4: $a\text{Cu} + b\text{HNO}_3 \rightarrow c\text{Cu}(\text{NO}_3)_2 + d\text{NO}_2 + e\text{H}_2\text{O}$

Cu: $a = c$ Let $a = 1$:

H: $b = 2e$ $a = 1$

N: $b = 2c + d$ $c = 1$

O: $3b = 6c + 2d + e$ $b = 2e \rightarrow e = b/2$

$b = 2 + d$

$a = 1, c = 1, b = 4, e = 2, d = 2$ $d = b - 2$

(if non-integer multiply $3b = 6 + 2(b - 2) + (b/2)$

accordingly so all integers) $0.5b = 2$ $e = 2$ and $d = 2$
 $b = 4$

Final balanced equation:



State Symbols (By default)

Solid (s)

- Metals except mercury
- Ionic compounds
- Giant covalent substances

Liquid (l)

- Mercury, Bromine
- Water, Ethanol

Aqueous (aq) - Dissolved in water

- Acids, Bases (when dissolved in water)
- Solutions

Gas (g)

- Simple covalent substances
- Gaseous elements $\rightarrow \text{H}_2, \text{N}_2, \text{O}_2, \text{F}_2, \text{Cl}_2$ + Noble gases

Write the state symbols after the molecule.

Example: $\text{H}_2\text{O}(\text{l})$

Writing Ionic Equations

Step 1: Write a balanced chemical equation including state symbols

Step 2: For ionic compounds in aqueous state, write their ions. The rest just copy down.

Step 3: Cancel out spectator ions. (same ions on LHS and RHS)

Step 4: Rewrite ionic equation.

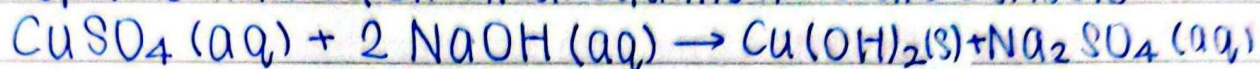
Check Answer: 1. Overall charges on both sides equal

2. Number of atoms on both sides equal

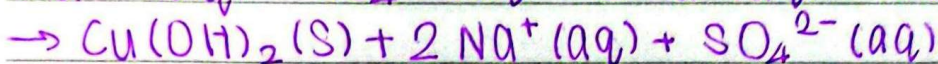
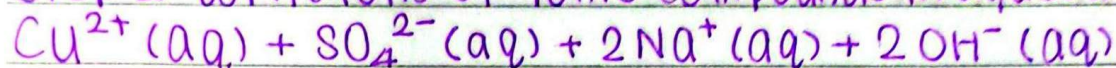
Example:

copper(II) sulphate + sodium hydroxide $\rightarrow$ copper(II) hydroxide
+ sodium sulfate

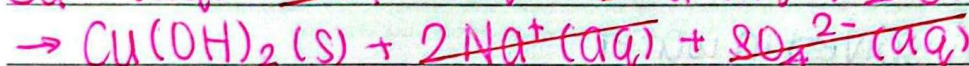
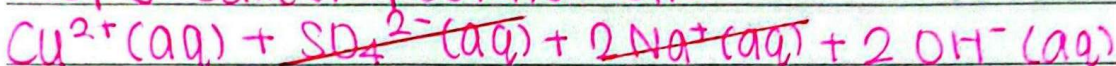
Step 1: Balanced chemical equation + state symbols



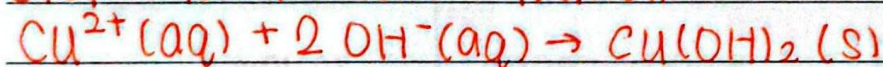
Step 2: Write ions of ionic compounds in aqueous state



Step 3: Cancel spectator ions.



Step 4: Rewrite ionic equation.



Check!

✓ Same overall charges on both sides.

✓ Same number of atoms on both sides.